

Application of Backward Stochastic Differential Equations to Modeling Financial Securities

Sofya Tsay

Supervisor: Yaroslav Lyulko, PhD

International College of Economics and Finance
Higher School of Economics
Moscow, Russia

20 June, 2026

Outline

1. Motivation
2. Mathematical foundations: BSDE and pricing
3. Numerical solvers
 - ▶ Least-squares Monte Carlo
 - ▶ Deep BSDE
4. Results
 - ▶ Validation on synthetic data
 - ▶ Application to real market data: five-asset basket
5. Conclusions and future work

Introduction

Motivation

- ▶ Fair pricing and hedging are among the most fundamental problems in quantitative finance.
- ▶ Classical methods address these two questions **separately**, and **one instrument type at a time**:
 - ▶ European — BSM formula
 - ▶ American — binomial trees
 - ▶ baskets — Monte Carlo
- ▶ No single framework unifies pricing and hedging across security types.

Novelty

- ▶ A **unified pricing engine** based on BSDE: European, American, and basket options in one numerical framework.
- ▶ The BSDE solution is a pair of processes: the engine outputs the **fair value** and the **delta hedge** simultaneously.
- ▶ The deep BSDE solver extends the engine to high dimensions and overcomes the **curse of dimensionality**.

Mathematical foundations

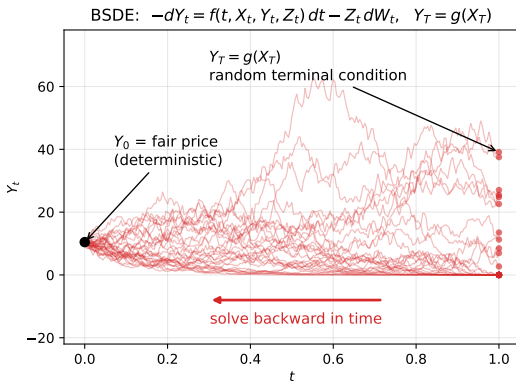
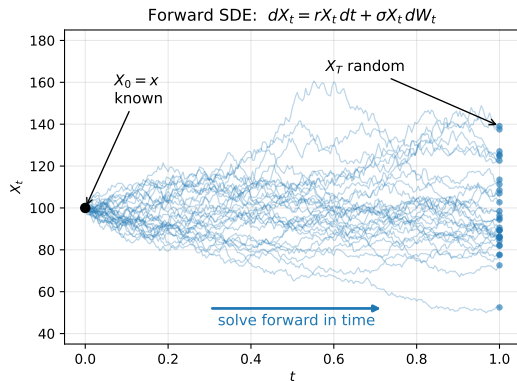
Definition

A solution of a *backward stochastic differential equation (BSDE)* is a pair of adapted processes (Y, Z) satisfying:

$$-dY_t = f(t, Y_t, Z_t) dt - Z_t^T dW_t, \quad Y_T = \xi, \quad (1)$$

where $\xi \in L^2(\mathcal{F}_T, P)$, and the function $f_s(y, z) = f_s(y, z; \omega)$ is $\mathcal{F}_s$ -measurable. Solution of a BSDE is a pair of adapted processes (Y, Z) . $\xi \in L^2(\mathcal{F}_T, P)$ is called terminal condition, and $f_s(y, z) = f_s(y, z; \omega)$ is called driver or generator.

Forward SDE vs Backward SDE



Pricing and hedging European options via BSDE

Hedging portfolio. Under the Black–Scholes–Merton model, a self-financing portfolio Y_t holding Δ_t units of stock, the remainder in the money-market account, satisfies

$$dY_t = rY_t dt + \underbrace{\Delta_t \sigma X_t}_{=: Z_t} dW_t, \quad Y_T = g(X_T).$$

This is a **linear BSDE**: the terminal condition is the option payoff, and the equation is solved backward in time.

Solution [El Karoui, Peng, Quenez, 1997]:

$$Y_t = \mathbb{E} \left[e^{-r(T-t)} g(X_T) \mid \mathcal{F}_t \right]$$

Y_t is the fair value of the option; $Z_t = \Delta_t \sigma X_t$ recovers the delta hedge.

Numerical solver I: least-squares Monte Carlo for BSDE

Discretising the BSDE and taking conditional expectations [Gobet, Lemor, Warin, 2005]:

$$Y_{t_i} = \mathbb{E} \left[Y_{t_{i+1}} + f(t_i, X_{t_i}, Y_{t_{i+1}}, Z_{t_i}) \Delta t \mid X_{t_i} \right], \quad Z_{t_i} = \frac{1}{\Delta t} \mathbb{E} \left[Y_{t_{i+1}} \Delta W_i \mid X_{t_i} \right].$$

LSMC [Longstaff, Schwartz, 2001]: regress on a basis of polynomials,

$$\mathbb{E} \left[V_{t_{i+1}} \mid X_{t_i} = x \right] \approx \sum_{k=1}^K \beta_{ik} \psi_k(x),$$

with coefficients β_i estimated by OLS across simulated paths.

Fast (a single pass in seconds), fully interpretable, and extends naturally to American options.

Extension to American options: reflected BSDE

American option can be exercised at any $t \in [0, T]$: the seller's portfolio must stay above the payoff, $Y_t \geq h(X_t)$ for all t .

Reflected BSDE. The solution becomes a triple (Y_t, Z_t, K_t) :

$$Y_t = \xi + \int_t^T f(s, Y_s, Z_s) ds + K_T - K_t - \int_t^T Z_s dW_s,$$

where K_t is increasing, $K_0 = 0$, and $\int_0^T (Y_t - h(X_t)) dK_t = 0$: the *minimal* upward push keeping Y above the obstacle. K acts only when $Y_t = h(X_t)$ — exactly where early exercise is optimal.

In the LSMC scheme this is one extra line per backward step:

$$Y_{t_i} = \max\left(\tilde{Y}_{t_i}, h(X_{t_i})\right)$$

The discretised reflected BSDE is algebraically equivalent to the Longstaff–Schwartz algorithm.

BSDE-LSMC on synthetic data: validation against benchmarks

Parameters: $S_0 = K = 100$, $r = 0.05$, $\sigma = 0.20$, $T = 1$; Hermite degree-4 basis.

Instrument	Benchmark	Price bias	Delta bias
European call	BSM formula	-0.04%	+0.09%
American put (ATM)	CRR tree	+0.14%	-0.85%
American put, $S_0 \in [85, 115]$	CRR tree	$< 0.22\%$	$< 2.2\%$
Basket call, $d = 3$	antithetic MC	-0.02%	$\leq 3.3\%$
Basket call, $d = 5$	antithetic MC	+0.20%	up to 7.9%

- **Prices:** no structural error for any instrument type — American put biases alternate in sign across moneyness (Monte Carlo noise, not bias); basket prices fall inside the MC 95% confidence interval.
- **Deltas:** accurate up to $d = 3$, but the degree-2 polynomial Z -basis degrades at $d = 5$ — the curse of dimensionality.

Numerical solver II: deep BSDE

Idea [Han, Jentzen, E, 2018]: reformulate the BSDE as a global optimisation problem solved *forward* in time.

The initial price $Y_0 = \theta_{Y_0}$ is a **learnable parameter**; at each time step a neural network approximates the hedge Z_{t_i} . Propagate forward,

$$Y_{t_{i+1}} \approx Y_{t_i} + r Y_{t_i} \Delta t + Z_{t_i}^\top \Delta W_i,$$

and train all parameters by minimising the terminal error

$$\ell(\theta) = \mathbb{E} \left| Y_T - g(X_T) \right|^2.$$

Such approach **overcomes the curse of dimensionality**, as neural networks are universal approximators — no polynomial degree constraint on $Z(t, X_t)$. However, forward propagation cannot handle early exercise.

Application to real market data: calibration

Five-asset basket call on large-cap US equities, evaluation date 19 May 2026; three years of daily data.

Asset	$S_0^{(j)}$ (USD)	$\hat{\sigma}_j$ (%)
AAPL	297.84	25.69
XOM	160.49	22.91
JPM	300.73	22.66
JNJ	228.92	17.25
AMZN	264.86	30.92

Real market data: BSDE-LSMC vs deep BSDE

Both solvers applied to the calibrated basket; benchmark: antithetic MC with $N = 500,000$ paths (95% CI half-width ± 0.0375).

Method	Price	Rel. error	Delta RMSE	Worst delta error
MC antithetic	19.7407	—	—	—
BSDE-LSMC	19.7506	+0.05%	0.00619	8.65% (XOM)
Deep BSDE	19.7559	+0.08%	0.00072	1.22% (JNJ)

- ▶ **Prices:** both solvers fall inside the MC 95% confidence interval — the engine is correct on real data, not only in controlled experiments.
- ▶ **Deltas:** the neural Z -approximation reduces delta RMSE by an order of magnitude — the degree-2 polynomial basis is the binding constraint at $d = 5$, exactly as the synthetic experiments predicted.

Conclusions

- ▶ BSDE provide a **genuinely unified pricing engine**: European, American, and basket options are priced and hedged within one framework, with the delta delivered as part of the solution.
- ▶ **BSDE-LSMC**: fast, fully interpretable, handles early exercise via reflection; price errors below 0.25% across all instruments and dimensions.
- ▶ **Deep BSDE**: overcomes the curse of dimensionality in delta estimation — RMSE falls from 0.00619 to 0.00072 at $d = 5$ — at the cost of training time and no American option support.
- ▶ The two solvers are **complementary**: LSMC when speed, interpretability, and early exercise matter; deep BSDE when accurate high-dimensional hedging is the priority.

Future work

- ▶ **Complex products:** Bermudan swaptions and autocallable notes.
- ▶ **Valuation adjustments (XVA):** CVA and DVA enter the replicating portfolio as a nonlinear BSDE driver.
- ▶ **Incomplete markets** [Shiryaev, 2025]: unknown drift with $r - k\sigma_t \leq \mu_t \leq r + k\sigma_t$ — upper and lower option prices solve BSDE with non-linear drivers, computable with the methods of this work; calibrating the market price of risk remains an open question.

Thank you for your attention!